

Semipalmated Sandpiper (SESA) Migration Stopover Site Analysis
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[Semipalmated Sandpiper Overview, All About Birds, Cornell Lab of Ornithology](#)

Problem and Hypothesis:

- SESA populations are on the decline and stopover habitat loss is believed to be a large part of this. Coastal wetlands are considered a primary habitat for stopover sites, and they are threatened by climate change and habitat changes.
- My main question: How have changes in coastal habitats (including sea level rise and habitat loss) negatively affected/ altered stopover sites that SESA use during migration?
- Hypothesis: SESA stopover sites (hotspots) along the US Atlantic COast have undergone significant coastal habitat change within the years 2001-2025. Stopover sites with more habitat changes and greater risk of sea level rise exposure will likely decline in stopover site usage over time.

Data and Preprocessing:

- 3 datasets
 - eBird Basic Dataset** — <https://ebird.org/data/download>
 - Citizen science bird observation records
 - Used to track SESA detections along the Atlantic coast
 - NLCD Annual Land Cover** — <https://www.mrlc.gov/data>
 - Satellite maps of land use
 - Used to track coastal habitat change
 - NOAA Sea-Level Rise Extent** — <https://coast.noaa.gov/slrdata/>
 - State level maps showing areas impacted by sea level rise (0.5m, 1m)
- Preprocessing steps:
 - Filter to only include migration months
 - April - May
 - August - October
 - Filtered to include 2001-2025 (that is when I could get land cover data from)
 - Final dataset: 276,883 detection records, 22,682 localities, 10 states
 - Converted to spatial object (using the CRS 4326 coordinate system) for mapping

Database: Postgres SQL

- Postgres SQL was chosen because I am comfortable using Postgres, it integrates well with R, and is good with handling large datasets.

Experiment Plan:

- NLCD land cover class, land cover changes, and sea level rise inundation will be predictor variables. SESA presence is going to be the response variable. The top 200 localities based on detections are serving as hotspots (stopover sites) and randomly generated background points are used as absence sites. I plan to use k-fold cross validation to evaluate but I realize there may be issues with this due to the spatial aspect of my data. Alternative methods such as spatial cross-validation may be used if needed.

ML Methods:

- Logistic Regression (Baseline)
 - To determine whether a site has SESA present or absent
 - What is the chance that given certain habitat (based on land cover class and slr vulnerability) characteristics SESA will use that location as a stopover site
 - Model performance evaluation metrics: accuracy, AUC, sensitivity, specificity
- Random Forest (will this improve from the baseline)
 - To capture nonlinear relationships between habitat characteristics and SESA presence that aren't captured in a linear model.
 - Model performance evaluation metrics: accuracy, AUC, variable importance scores

AI Use

- Claude was used for a few aspects of my project thus far. I used it as a tool to assist in the direction of my project. I had a broad idea of what I wanted to do but used it to help me narrow down my topic and goal to something achievable and meaningful. I also used it to explain some R packages that were new to me, to help me with code when working with the spatial data formats and with some formatting of my visuals (centering titles, etc).